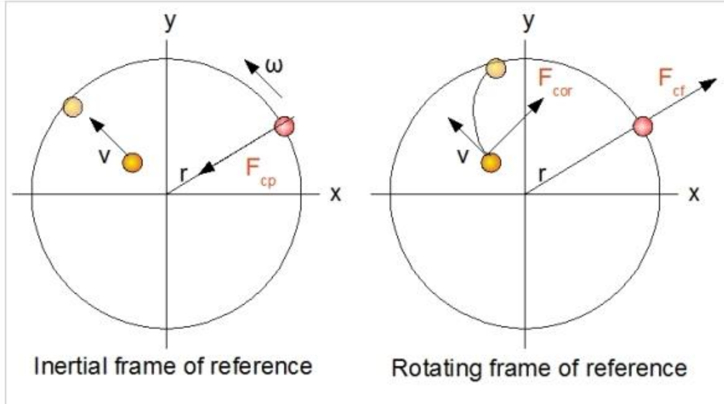
Manual to Lab 4: PHY2048C

Florida State University

Rotating Frames



About labs in this class

The labs in this class will have general instructions, and many things need to be figured out by the students. **Answer the questions and record your measurements in your lab notebook and then submit the notebook at the end of the activity.**

About this lab

In this lab, you will measure the elastic constant of a spring by considering a rotating frame and a centrifugal force. You will also estimate the torque of an electric motor.

Activity 1. Turn on the device. Draw a diagram explaining its workings. Get the spring to expand. When the mass is fully extended, a needle below the spring goes up. Make it so that the ring is extended and this needle remains lowered.

Activity 2. Draw a free body diagram of the mass of the previous situation on an inertial frame. Assume there is no friction (the actual friction is very small).

Activity 3. Draw a free body diagram of the mass attached to the spring on a rotating frame centered at the spring's center of rotation.

Question 1: What other forces arise when considering a rotating frame?

Activity 4: Determine the elastic constant k.

Activity 5: Compare your value with the other three groups and make a histogram of the four values. Use the standard deviation as the error in .

Question 2: Estimate the torque that the electric motor exerts on the belt. Assume the whole device is made of iron ($\rho = 7.87\frac{g}{cm^3}$) and that the belt is massless. (Hint: $\tau \approx \frac{\Delta L}{\Delta t}$)

Question 3: Now use this to find the power delivered the motor and compare this to the reported power given by the manufacturer. Find the discrepancy. If the discrepancy is high (more than 100%), why? If it isn't even though you ignored the belt, how come this assumption worked? Is the measured value higher or lower? why?